

NIIS HACKATHON 2026 — OFFICIAL PROBLEM BRIEF

Track: Campus Nexus

Challenge ID: PS-02

Title: CAMPUSLINK – AI-Powered Campus-to-Corporate Placement Management

1. Executive Summary & Problem Overview

Every placement season, college Training & Placement (T&P) cells manage thousands of graduating candidates, hundreds of corporate recruiters, and dozens of overlapping hiring drives. Currently, this critical transition relies on fragile spreadsheets, uncoordinated WhatsApp groups, and fragmented email threads—leading to schedule collisions, communication lapses, and opaque shortlisting criteria.

Students lack continuous readiness profiling and clear visibility into role-specific skill gaps before facing technical interviews. Concurrently, corporate recruiters struggle with uncured applicant pools and rigid filtering that discards high-potential candidates.

The objective of **PS-02 (CAMPUSLINK)** is to design and build an intelligent, data-driven Campus-to-Corporate Placement Management System that automates candidate profiling, performs explainable AI-driven matchmaking, executes collision-free drive scheduling, and delivers near real-time predictive placement analytics.

2. Core Personas & Access Roles

1. **Student / Candidate:** Maintains a dynamic readiness profile, uploads verifiable projects and credentials, views role-specific skill gap diagnostics, applies to eligible drives, and tracks interview/offer pipelines.
2. **Corporate Recruiter:** Posts Job Descriptions (JDs), defines granular eligibility rubrics (skills, cutoffs, certifications), reviews pre-screened candidate pools with transparent fit scores, and manages drive stages.
3. **Training & Placement Officer (TPO) / Admin:** Oversees campus-wide recruitment schedules, detects and resolves venue/time conflicts, tracks batch-wide conversion rates, and flags at-risk students for remedial training.
4. **Faculty Mentor / Interviewer:** Conducts mock interviews, inputs standardized soft-skill and technical evaluations, and reviews student progression analytics.

3. Key Functional Modules & Technical Deliverables

Module 1: Dynamic Student Readiness & Skill-Gap Engine

- **Multimodal Profile Extraction:** Parses resumes (PDF), GitHub repositories, academic transcripts (CGPA, backlog history), and certifications.
- **Employability Scoring:** Assigns an explainable readiness index: $\text{Readiness Level} \in \{\text{Not Ready} \rightarrow \text{Developing} \rightarrow \text{Ready} \rightarrow \text{Highly Employable}\}$
- **Personalized Skill Gap Diagnostic:** Directly flags missing competencies against target corporate profiles (e.g., "*Candidate meets CGPA criteria but lacks cloud deployment experience required by JD*").

Module 2: Explainable Recruiter-Candidate Matching

- **Semantic Job Description Parser:** Extracts mandatory technical stacks, soft-skill benchmarks, and minimum eligibility criteria.
- **Match Scoring Algorithm:** Generates a multidimensional fit percentage (0–100%) factoring in project relevance, mock-interview ratings, and core competencies.
- **Transparent Justification:** Replaces black-box "*Not Shortlisted*" rejections with transparent inclusion/exclusion audit logs for candidates and recruiters.

Module 3: Conflict-Aware Drive & Interview Scheduler

- **Smart Collision Prevention:** Prevents double-booking of physical campus venues (auditoriums, labs) and candidate interview time slots across overlapping corporate drives.
- **Stage Tracking & Pipeline Automation:** Tracks student status through aptitude tests, technical rounds, HR evaluations, offer letters, and joining verification.
- **Automated Notification Dispatch:** Trigger automated notifications and reminders via Email/WhatsApp/SMS for drive deadlines, shortlist reveals, and document submissions.

Module 4: Predictive Placement Analytics Dashboard

- **Conversion Rate Analytics:** Real-time visibility into branch-wise conversions, recruiter return frequency, and median CTC trends.
- **At-Risk Cohort Detection:** Early algorithmic identification of students consistently underperforming in preliminary rounds to trigger targeted mentor interventions.
- **Skill Market Demands:** Insights into emerging industry hiring trends to assist faculty in tuning academic elective offerings.

4. Recommended Technology Stack

Layer, Suggested Technologies

Frontend Web, React.js (Vite) / Next.js / Tailwind CSS

Backend Architecture, Node.js (Express) / Python (FastAPI / Django)

Database, PostgreSQL / MongoDB / Supabase / Firebase Firestore

AI / NLP & Recommendation,"Sentence-Transformers, OpenAI API / Gemini API, spaCy, LangChain, FAISS / Pinecone (for vector resume search)"
Scheduling Engine,"Interval scheduling algorithms, Google Calendar API integration, BullMQ / Redis for automated reminders"
Authentication & RBAC,"JWT, OAuth2, Role-Based Access Control"

5. Evaluation & Judging Criteria (100 Points Total)

1. **Readiness Profiling & Skill-Gap Analysis (25 Points):** Accuracy in resume parsing, completeness of the student readiness index, and actionable skill-gap recommendations.
2. **Matching Engine & Explainability (25 Points):** Quality of candidate-to-JD recommendation and transparency of match justifications.
3. **Collision-Free Scheduling & Workflow Automation (20 Points):** Real-time conflict handling, multi-stage drive pipeline management, and automated notifications.
4. **Analytics & Predictive Insights (20 Points):** Quality of placement cell intelligence, at-risk student escalation, and institutional trend forecasting.
5. **System Architecture, 24h Sprint Prototype & Demo (10 Points):** Working multi-role prototype, code quality, and repository documentation.

6. Submission Guidelines

- **Code Repository:** Public/Private GitHub repository containing clean branch commits, setup guide, and .env.example.
- **Live Deployment:** Hosted interactive web portal (Vercel / Render / AWS).
- **Demo Video / Deck:** A structured 3-minute video demonstrating end-to-end recruiter JD posting, candidate matching, and conflict-free interview scheduling.